Data Analysis and Visualization Using Python

ARIVANI TECHNOLOGIES PVT LTD

A Summer Training Report

Submitted in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications (MCA)

By

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Centre for Distance and Online Education

LOVELY PROFESSIONAL UNIVERSITY

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ANNEXURE-II: STUDENT DECLARATION

To whomsoever it may concern

I, Kaif Hashmi, Registration Number [REGISTRATION NUMBER], hereby declare that the work presented in this Summer Training Report titled “Data Analysis and Visualization Using Python” was carried out by me as part of the requirements for the Master of Computer Applications (MCA) programme.

The project focuses on data cleaning, data processing, exploratory analysis, visualization, statistical testing and basic forecasting using Python. The dataset used in the practical analysis is simulated retail-sales data, as stated in the project presentation and report, and is not presented as confidential company data.

I further declare that the report is prepared for academic purposes and that external sources used in the work have been acknowledged through appropriate references.

Name of the Student: Kaif Hashmi ([REGISTRATION NUMBER])

Signature of the Student: _____

Dated: _____

ANNEXURE-III: SUMMER TRAINING COMPLETION CERTIFICATE

[ORIGINAL ORGANIZATION CERTIFICATE TO BE INSERTED HERE]

This page is reserved for the official Summer Training Completion Certificate issued by the organization. The LPU guideline requires the certificate to identify the student, project title, training period, supervising person and attendance details. The original signed and stamped certificate should be inserted in the final submission.

Organization: Arivani Technologies PVT LTD

Student Name: Kaif Hashmi

Project Title: Data Analysis and Visualization Using Python

Industry Mentor: [VERIFY AND INSERT OFFICIAL NAME]

Training Period: [VERIFY AND INSERT DATES]

Attendance: [VERIFY AND INSERT OFFICIAL ATTENDANCE]

Authorized Signatory: _____

Organization Seal: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to Lovely Professional University and the Centre for Distance and Online Education for providing the academic framework for summer training. This training gave me an opportunity to connect classroom knowledge with practical data-analysis work.

I am thankful to the organization and the training team for providing an environment in which I could study data analysis, Python libraries, data cleaning, visualization and statistical interpretation. I also appreciate the guidance received during the practical work and documentation process.

I am grateful to my faculty members and mentors for their guidance and feedback. Finally, I thank everyone who supported me in completing the project and preparing the final report.

ABSTRACT

This Summer Training project, “Data Analysis and Visualization Using Python”, demonstrates a complete and simple data-analysis workflow using Python. The project uses a simulated retail-sales dataset so that the complete analytical process can be practiced without using confidential business information.

The work begins with generating and loading data, checking its structure and identifying quality problems. The raw dataset contains 5,060 rows and 13 columns. The project identifies 52 duplicate rows, 220 missing cells and 25 quantity outliers. The data is then cleaned by removing duplicates, standardizing text, filling selected missing numeric values with the median and capping extreme quantity values using the IQR method. After cleaning, 5,008 records remain.

The cleaned data is used to calculate key performance indicators, study monthly revenue, compare product categories and regions, analyze customer age groups and payment modes, examine correlations, perform statistical tests and prepare a simple three-month revenue forecast. Python libraries including Pandas, NumPy, Matplotlib, Seaborn and SciPy are used throughout the workflow.

The project shows how raw data can be converted into understandable information. It also develops practical skills in data preparation, exploratory data analysis, visualization, statistical reasoning, reproducibility and technical reporting.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
MCA	Master of Computer Applications
LPU	Lovely Professional University
CDOE	Centre for Distance and Online Education
STS	Summer Training Seminar
EDA	Exploratory Data Analysis
KPI	Key Performance Indicator
AOV	Average Order Value
IQR	Inter-Quartile Range
CSV	Comma-Separated Values
PNG	Portable Network Graphics
UPI	Unified Payments Interface
COD	Cash on Delivery
ANOVA	Analysis of Variance
API	Application Programming Interface
APA	American Psychological Association

Chapter-1: INTRODUCTION TO THE PROJECT

1.1 Introduction to Summer Training

Summer training provides an opportunity to connect academic learning with practical work. The LPU guideline explains that training helps students apply knowledge, understand workplace practices and develop technical or managerial skills. For the online programme, the guideline specifies a six-week summer training course.

1.2 Background of the Project

Organizations create large amounts of data through sales, customers, payments, delivery activities and feedback. Simply storing this data is not enough. It must be checked, cleaned and analyzed before it can support useful decisions. This project was selected to demonstrate that complete process in a simple and understandable way.

1.3 Project Identification

The project is titled “Data Analysis and Visualization Using Python”. Its practical focus is retail-sales data. The project demonstrates how Python can be used to move from raw records to meaningful summaries, charts, statistical results and a basic forecast.

1.4 Problem Statement

Raw datasets often contain duplicate records, missing values, inconsistent text and unusual observations. If such problems are ignored, the results of analysis may be misleading. The problem addressed by this training is therefore to build a reliable process that cleans the data first and then produces clear, evidence-based findings.

1.5 Objectives of the Work Undertaken

No.	**Objective**
1	Understand the structure and quality of a retail-style dataset.

2	Identify and correct missing values, duplicates, text inconsistencies and outliers.
3	Create useful features such as revenue, month, weekday, weekend flag and age group.
4	Calculate important business indicators and grouped summaries.
5	Create charts that communicate trends and comparisons clearly.
6	Apply basic statistical tests and correlation analysis.
7	Prepare a simple revenue forecast and document the complete workflow.

1.6 Scope of the Work

The scope covers data generation and loading, data inspection, cleaning, feature engineering, descriptive analysis, KPI calculation, visualization, statistical testing and basic forecasting. The project does not claim to be a production forecasting system or a confidential organizational analytics system. The dataset is simulated for academic training.

1.7 Importance and Applicability

The project is useful because the same workflow is applicable to many real datasets. A business can use similar steps to understand sales performance, customer behavior, payment preferences, delivery performance and product demand. The techniques are also useful for dashboards, reporting and later machine-learning projects.

1.8 Relevance of Python

Python is suitable for this work because it provides readable syntax and a strong data ecosystem. Pandas simplifies table operations, NumPy supports numerical calculations, Matplotlib and Seaborn provide visualizations, and SciPy provides statistical methods. These libraries allow the complete workflow to be performed in one environment.

1.9 Role and Profile of the Student

During the training, the student worked as a learner and analyst. The main responsibilities were understanding the project requirement, preparing the dataset, writing and checking Python code, interpreting outputs, creating visualizations, documenting results and preparing the presentation.

1.10 Training Work Plan

Week	**Main Work**
Week 1	Project understanding, Python

	environment and data-analysis basics.
Week 2	Dataset generation, loading and initial inspection.
Week 3	Data cleaning, missing-value treatment, duplicate removal and outlier handling.
Week 4	Feature engineering, KPI calculation and exploratory analysis.
Week 5	Visualization, correlation, statistical tests and forecasting.
Week 6	Interpretation, documentation, presentation and final review.

1.11 Expected Learning Outcomes

• Ability to inspect and clean tabular data.

• Ability to use Pandas and NumPy for data processing.

• Ability to create useful charts with Matplotlib and Seaborn.

• Ability to understand basic statistical results.

• Ability to communicate analytical findings in simple language.

• Ability to prepare reproducible project evidence and documentation.

Chapter-2: INTRODUCTION OF THE COMPANY / WORK

2.1 Organization / Training Context

The training context for this project is Arivani Technologies PVT LTD. The project work is focused on Python-based data analysis and visualization. Organization-specific facts that are not verified in the available training evidence are intentionally not added to this report.

2.2 Nature of the Training Work

The work is technical and analytical. It combines Python programming with data handling and communication. The practical output includes cleaned CSV data, analytical summaries, charts, statistical results, a forecast and academic documentation.

2.3 Training Role and Activities

Activity	**Tool / Output**
Dataset generation and loading	NumPy + Pandas + CSV
Data cleaning	Pandas + cleaned CSV
Statistical analysis	SciPy + analytical results
Visualization	Matplotlib + Seaborn + PNG charts
Documentation	Word/PDF report
Presentation	PowerPoint presentation

2.4 Work Approach

The work followed a step-by-step approach rather than creating charts directly from raw data. First the dataset was inspected. Next the quality problems were identified and corrected. After cleaning, new fields were created and the data was summarized. Only then were charts and statistical tests prepared. This order reduces the chance of drawing conclusions from poor-quality records.

2.5 Professional Relevance

The project develops skills that are useful for entry-level data analyst, Python developer and software development roles. It also provides a foundation for later work in business intelligence, machine learning, automation and dashboard development.

Chapter-3: PYTHON TOOLS AND ANALYTICAL WORKFLOW

3.1 Introduction to Data Analysis

Data analysis is the process of examining data to answer questions and identify useful patterns. A good analysis normally includes data preparation, descriptive summaries, comparison, visualization and interpretation. The result should be understandable to a person who did not write the code.

3.2 Data Visualization

Data visualization converts numbers into visual forms such as line charts, bar charts, histograms, box plots and heatmaps. In this project, charts are used to show revenue trends, category differences, regional performance, customer segments, payment shares and relationships between variables.

3.3 Python Libraries Used

Library	**Main Use**
Pandas	Read CSV files, clean records, filter data, group data and create summaries.
NumPy	Generate reproducible data and perform numerical operations.
Matplotlib	Create line, bar and other basic charts.
Seaborn	Create statistical charts and heatmaps.
SciPy	Perform t-test, ANOVA, Pearson correlation and regression-related analysis.

3.4 Analytical Workflow

Stage	Purpose	Output
Generate	Create simulated retail data	Raw CSV
Load	Read the dataset into Python	DataFrame
Inspect	Check shape, types, missing values and duplicates	Data-quality summary
Clean	Correct known data-quality problems	Clean DataFrame / CSV
Feature Engineering	Create revenue, time and customer features	Analysis-ready fields
Analyse	Calculate KPIs and grouped summaries	Tables and metrics
Visualize	Show trends and relationships	Charts
Test	Check relationships and differences statistically	p-values and coefficients
Forecast	Estimate short-term trend	Three-month forecast
Report	Explain methods and findings	Report and presentation

3.5 Reproducibility

A fixed NumPy random seed of 42 is used for the simulated dataset. The analytical workflow is organized into clear stages and the same results are used for the report and presentation. Reproducibility is

important because another person should be able to run the code and understand how the reported numbers were produced.

3.6 Python Environment

The practical environment can be a Python installation with Jupyter Notebook or an IDE such as Visual Studio Code. The required libraries are installed before execution. A simple requirements file can contain pandas, numpy, matplotlib, seaborn, scipy and scikit-learn when forecasting or regression utilities are used.

```
pip install pandas numpy matplotlib seaborn scipy scikit-learn
```

3.7 Importing the Main Libraries

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from scipy.stats import ttest_ind, f_oneway, pearsonr
```

3.8 Loading the Dataset

```
df = pd.read_csv("retail_sales_2024_raw.csv")

print(df.head())

print(df.shape)

print(df.info())
```

3.9 Why These Steps Matter

The first inspection prevents incorrect assumptions. Shape tells us how many records and fields exist. Info() shows data types and non-null counts. Head() gives a quick view of the actual values. These simple checks are important before any cleaning or statistical calculation is started.

Chapter-4: DATA COLLECTION, CLEANING AND PREPARATION

4.1 Data Collection

The dataset used in the project is simulated retail-sales data for the year 2024. It was generated with NumPy using seed 42 and contains approximately five thousand orders. The raw dataset has 13 columns. The simulated nature is clearly stated so that the project does not imply access to confidential company records.

4.2 Dataset Data Dictionary

Column	**Meaning**	**Type**
order_id	Order identifier	Text
order_date	Order date	Date
customer_id	Customer identifier	Text
customer_age	Customer age	Numeric
region	Sales region	Categorical
category	Product category	Categorical
product	Product name	Categorical
quantity	Units ordered	Numeric
unit_price	Price per unit	Numeric
discount	Discount percentage	Numeric
payment_mode	Payment method	Categorical
delivery_days	Delivery time	Numeric
rating	Customer rating	Numeric

4.3 Data Quality Problems

Problem	**Observed**	**Treatment**
Raw rows	5,060	Starting dataset

Duplicate rows	52	Removed using exact-row comparison
Missing cells	220	Median imputation for selected numeric fields
Quantity outliers	25	Capped using IQR rule
Text inconsistencies	Region/payment values	Standardized text
Clean rows	5,008	Final analysis dataset

4.4 Why Cleaning Is Necessary

Duplicate records can increase totals incorrectly. Missing values can cause calculations to fail or reduce the available sample. Inconsistent text creates separate groups for what should be the same category. Extreme quantities can distort averages and charts. Cleaning therefore improves the reliability of later analysis.

4.5 Removing Duplicate Records

```
df = df.drop_duplicates()

print("Rows after duplicate removal:", len(df))
```

4.6 Standardizing Text

```
df["region"] = df["region"].astype(str).str.strip().str.title()

df["payment_mode"] = df["payment_mode"].astype(str).str.strip().str.title()
```

4.7 Handling Missing Values

```
df["customer_age"] = df["customer_age"].fillna(df["customer_age"].median())

df["delivery_days"] = df["delivery_days"].fillna(df["delivery_days"].median())

df["rating"] = df["rating"].fillna(df["rating"].median())
```

4.8 Handling Quantity Outliers

```
q1 = df["quantity"].quantile(0.25)

q3 = df["quantity"].quantile(0.75)

iqr = q3 - q1

upper = q3 + 1.5 * iqr

df["quantity"] = df["quantity"].clip(upper=upper)
```

4.9 Checking the Clean Dataset

```
print("Duplicates:", df.duplicated().sum())
print("Missing values:", df.isna().sum().sum())
print(df.shape)
```

The expected cleaned result is 5,008 rows with the identified missing values handled and duplicate rows removed. The cleaned dataset is then suitable for grouped analysis, distributions, correlations, hypothesis tests and monthly trend analysis.

4.10 Feature Engineering

Feature engineering means creating new fields from existing fields so that the data can answer more useful questions. In this project, revenue and time-related fields are created from the order information.

```
df["order_date"] = pd.to_datetime(df["order_date"])
df["gross_amount"] = df["quantity"] * df["unit_price"]
df["revenue"] = df["gross_amount"] * (1 - df["discount"] / 100)
df["month"] = df["order_date"].dt.month
df["weekday"] = df["order_date"].dt.day_name()
df["is_weekend"] = df["weekday"].isin(["Saturday", "Sunday"])
```

4.11 Creating Age Groups

```
bins = [0, 25, 35, 45, 55, 100]
labels = ["18–25", "26–35", "36–45", "46–55", "56+"]
df["age_group"] = pd.cut(df["customer_age"], bins=bins, labels=labels)
```

4.12 Saving the Clean Data

```
df.to_csv("retail_sales_2024_cleaned.csv", index=False)
```

4.13 Data Preparation Result

The result of preparation is a consistent analytical table. The dataset now has standardized categorical values, treated missing values,

controlled quantity extremes and additional fields that support monthly, customer, category and regional analysis.

Chapter-5: EXPLORATORY ANALYSIS AND VISUALIZATION

5.1 Key Performance Indicators

KPI	Value
Total Revenue	169,384,561.26
Gross Sales	202,063,565.07
Total Orders	5,008
Unique Customers	1,704
Average Order Value	33,822.80
Average Rating	3.97 / 5
Average Delivery Days	3.51
Average Discount	16.33%
Total Quantity	15,263

These indicators provide a quick overview of the dataset. Revenue and gross sales show the financial scale, orders and customers show activity, AOV shows typical order size, while rating, delivery days and discount describe customer experience and pricing behavior.

5.2 Calculating KPIs in Python

```
total_revenue = df["revenue"].sum()

total_orders = len(df)

unique_customers = df["customer_id"].nunique()

avg_order_value = df["revenue"].mean()

avg_rating = df["rating"].mean()
```

5.3 Monthly Revenue Trend

Monthly analysis helps identify periods of stronger and weaker sales. The reference presentation uses a monthly revenue chart to

communicate the trend clearly. The trend is especially useful for planning inventory, staffing and marketing activity.

```
monthly_revenue =
df.groupby(df["order_date"].dt.month)
["revenue"].sum()

monthly_revenue.plot(kind="line", marker="o")

plt.title("Monthly Revenue Trend")

plt.xlabel("Month")

plt.ylabel("Revenue")

plt.show()
```

5.4 Revenue by Product Category

Category analysis compares the contribution of different product groups. In the project, Electronics is the dominant category and contributes 76.9% of total revenue. This concentration means that electronics performance has a major effect on the overall result.

```
category_revenue = (df.groupby("category")
["revenue"].sum()
                    .sort_values(ascending=F
alse))
category_revenue.plot(kind="bar")
plt.title("Revenue by Product Category")
plt.ylabel("Revenue")
plt.show()
```

5.5 Revenue by Region

Regional analysis shows where revenue is generated. North, West and South together contribute 72.6% of total revenue. The remaining regions provide opportunities for growth and closer investigation.

```
region_revenue = (df.groupby("region")
["revenue"].sum()
                  .sort_values(ascending=Fal
se))
```

5.6 Revenue by Customer Age Group

Age-group analysis helps identify customer segments that contribute more revenue. Customers in the 26–45 range contribute about two thirds of revenue in the project data. This suggests that this group is important for targeted marketing and product communication.

5.7 Payment Mode Analysis

Payment mode analysis shows how customers complete purchases. UPI is the leading payment method with 38.5% of orders, followed by Cash on Delivery at 19.9%. This information can help prioritize payment experience and operational support.

```
payment_share = (df["payment_mode"].value_counts(normalize=True) * 100)
print(payment_share)
```

5.8 Customer Rating Analysis

The average customer rating is 3.97 out of 5. Ratings can be studied through a histogram or box plot. Low-rating orders should be investigated because they may indicate product, delivery or service problems even when the overall average looks acceptable.

```
sns.histplot(df["rating"], bins=5, kde=True)
plt.title("Distribution of Customer Ratings")
plt.show()
```

5.9 Correlation Analysis

Correlation measures the direction and strength of a linear relationship between two numeric variables. The project reports a strong positive relationship between unit price and revenue (r = 0.829). This is expected because revenue is directly influenced by unit price. Discount and quantity show a weak positive relationship (r = 0.127).

```
corr = df.select_dtypes("number").corr()
sns.heatmap(corr, annot=True, cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()
```

5.10 Discount and Quantity

The project compares average quantity across discount levels. The reference presentation reports that a 20% discount is associated with about 25% more units per order than no discount, while the increase does not continue at 25%. This suggests that a higher discount is not automatically better.

```
discount_quantity = df.groupby("discount")["quantity"].mean()
discount_quantity.plot(kind="bar")
plt.title("Average Quantity by Discount Level")
plt.show()
```

5.11 Statistical Testing

Statistical tests help determine whether an observed difference is likely to be meaningful or could have occurred by chance. A p-value below 0.05 is commonly treated as evidence against the null hypothesis at the 5% level. In this project, the reported tests are not statistically significant.

Test	Statistic	p-value	Interpretation
Weekend vs weekday order value	t = -0.342	0.7324	Not significant
Rating across regions	F = 1.723	0.142	Not significant
Delivery days vs rating	r = -0.011	0.4416	Not significant

5.12 Statistical Test Code

```
weekend = df.loc[df["is_weekend"], "revenue"]
```

```
weekday = df.loc[~df["is_weekend"], "revenue"]

t_stat, p_value = ttest_ind(weekend, weekday, equal_var=False)

r, p = pearsonr(df["delivery_days"], df["rating"])
```

5.13 Revenue Forecast

A simple linear trend is used to demonstrate forecasting. The reference presentation reports an increase of approximately ₹150,000 per month, $R^2 = 0.646$ and $p = 0.0016$. The indicative forecasts for months 13, 14 and 15 are approximately ₹4.84 million, ₹4.99 million and ₹5.14 million.

This forecast should be treated as a simple academic demonstration rather than a production forecasting model. It does not fully capture seasonal effects. A future version could use SARIMA, Prophet or another time-series method after collecting more historical data.

```
from scipy.stats import linregress

monthly = df.groupby(df["order_date"].dt.month)["revenue"].sum()

x = np.arange(1, len(monthly) + 1)

result = linregress(x, monthly.values)

forecast_x = np.arange(len(monthly) + 1, len(monthly) + 4)

forecast = result.intercept + result.slope * forecast_x
```

5.14 Overall Interpretation

The analysis shows that the project data is concentrated in a few important segments. Electronics are the largest category, the North/West/South regions account for most revenue, and the 26–45 customer group contributes strongly. UPI is the most common payment mode. Statistical tests do not show strong evidence for differences in weekend order value, regional ratings or the relationship between delivery time and rating.

5.15 Recommendations from the Analysis

• Prepare inventory and staff capacity for stronger sales periods.

• Monitor Electronics closely because of its large revenue contribution.

• Use targeted marketing for the 26–45 customer group while testing opportunities in other segments.

• Keep routine discounts around the range that shows useful quantity improvement rather than assuming higher discounts always produce better results.

• Investigate low customer ratings to identify product and service issues.

• Study lower-contributing regions for possible growth opportunities.

Chapter-6: TRAINING EXPERIENCE, CHALLENGES AND LEARNING OUTCOMES

6.1 Training Activities

The training activities were organized around the complete analytics lifecycle. The work included understanding the project, preparing the Python environment, generating and loading data, inspecting data quality, cleaning records, engineering features, calculating KPIs, creating charts, applying statistical tests, preparing a forecast and documenting the results.

6.2 Practical Work Completed

Work Area	**Practical Output**
Data preparation	Raw and cleaned CSV datasets
Data quality	Duplicate, missing-value and outlier checks
Feature engineering	Revenue, month, weekday, weekend and age group
Exploratory analysis	KPI and grouped summaries
Visualization	Charts for trends, categories, regions, age, payment and relationships
Statistics	t-test, ANOVA and Pearson correlation
Forecasting	Simple linear trend and three-month estimate
Documentation	Report and presentation

6.3 Challenges Faced

Challenge	**Solution**

Missing values	Inspect affected columns and use median imputation for selected numeric fields.
Duplicate rows	Use exact duplicate detection and remove repeated records.
Text inconsistency	Strip spaces and standardize categorical text.
Outliers	Use the IQR rule to identify and cap extreme quantities.
Interpreting charts	Focus on one question per chart and explain the result in simple language.
Statistical interpretation	Use p-values and correlation coefficients rather than relying only on visual differences.
Reproducibility	Use a fixed random seed and consistent source metrics.

6.4 Problem-Solving Experience

The project improved problem-solving because every unexpected data issue required a clear response. Instead of deleting information without checking, the process first identified the problem and then selected a suitable treatment. This approach is useful in professional analytics because data quality decisions can affect business conclusions.

6.5 Technical Knowledge Gained

• Working with DataFrames and CSV files using Pandas.

• Using NumPy for numerical operations and reproducible data generation.

• Cleaning missing values, duplicates, inconsistent text and outliers.

• Creating derived fields from dates and numeric columns.

• Grouping data and calculating summaries.

• Creating charts with Matplotlib and Seaborn.

• Using t-tests, ANOVA and Pearson correlation.

• Understanding a basic regression trend and forecast.

• Writing reproducible technical documentation.

6.6 Learning Outcomes

At the end of the training, the student is able to explain the basic data-analysis lifecycle, identify common data-quality problems, clean a tabular dataset, calculate meaningful indicators, select suitable charts, interpret simple statistical results and communicate findings clearly. The training also strengthens Python programming and debugging skills.

6.7 Training Effectiveness

The training is effective because it connects individual Python functions to a complete practical project. The student does not only learn how to call a library function; the work explains why the function

is required, what problem it solves and how the output supports the final conclusion.

6.8 Professional Skills

The project also develops documentation, communication, structured problem solving and presentation skills. These skills are important because a good analyst must explain results to both technical and non-technical audiences.

6.9 Evidence of Work

The main evidence produced by the project includes the raw simulated CSV, cleaned CSV, analytical metrics, Python code, charts, statistical results, forecast output, report and presentation. Keeping these outputs together creates a clear evidence trail from source data to final conclusion.

6.10 Reproducibility and Documentation

The project uses a fixed random seed and a defined workflow. The same analytical logic is reflected in the report and presentation. This reduces accidental differences between the documents and makes the project easier to review or reproduce.

6.11 Relation to MCA Learning

The project applies programming, data structures, mathematical reasoning, database-like tabular operations, software tools and technical communication. It therefore provides a practical connection between MCA academic learning and an industry-relevant data task.

6.12 Overall Training Experience

The overall experience provided practical exposure to a complete analytics task. The most important learning was that useful visualization depends on reliable data preparation and clear interpretation. The project also created a foundation for more advanced learning in SQL, dashboards, machine learning, web-based analytics and cloud data platforms.

Chapter-7: CONCLUSION

7.1 Introduction

This chapter summarizes the project and explains how the practical work achieved the objectives defined in Chapter-1.

7.2 Summary of the Training

The training covered the complete process of analyzing a simulated retail-sales dataset with Python. The work started with dataset generation and inspection, followed by cleaning, feature engineering, exploratory analysis, visualization, statistical testing and forecasting. The results were then organized into a report and presentation.

7.3 Achievement of Objectives

Objective	**Achievement**
Understand data quality	Raw structure and quality problems were identified.
Clean the dataset	Duplicates, missing values, inconsistent text and quantity outliers were treated.
Create features	Revenue, month, weekday, weekend flag and age group were created.
Calculate KPIs	Revenue, orders, customers, AOV, rating, delivery and discount metrics were calculated.
Visualize results	Multiple charts were prepared for trends, categories, regions, customers and relationships.
Apply statistics	t-test, ANOVA and Pearson correlation were used.

Forecast and communicate	A simple three-month forecast and complete documentation were prepared.

7.4 Key Findings

• Total revenue is ₹169,384,561.26 from 5,008 orders.

• Electronics contribute 76.9% of revenue.

• North, West and South contribute 72.6% of revenue together.

• Customers aged 26–45 contribute about two thirds of revenue.

• UPI accounts for 38.5% of orders.

• The unit price–revenue relationship is strong (r = 0.829).

• The tested weekend, regional rating and delivery-rating relationships are not statistically significant.

7.5 Conclusion

The project demonstrates that Python can be used to turn raw tabular data into useful information through a clear and repeatable process. The strongest lesson is that analysis should begin with data quality. Once the data is reliable, summaries and visualizations become more meaningful and statistical tests can be interpreted with greater confidence.

7.6 Future Scope

The project can be extended by using real organization-approved data, adding profit and cost measures, connecting the analysis to SQL databases, building an interactive dashboard, using advanced time-series forecasting, applying customer segmentation and developing machine-learning models for demand or customer behavior prediction.

7.7 Recommendations

• Continue practicing Pandas and SQL together for larger datasets.

• Learn dashboard tools such as Power BI or Tableau.

• Study advanced time-series methods after understanding the basic regression approach.

• Add automated data-quality checks to future projects.

• Use version control and clear project documentation for reproducibility.

• Validate business assumptions with domain experts when real company data is available.

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Note: Web references are listed in APA-style form and include access dates as required by the LPU summer-training guideline.

APPENDIX A: SELECTED PYTHON CODE

A.1 Complete Analytical Flow

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind, f_oneway, pearsonr, linregress

df = pd.read_csv("retail_sales_2024_raw.csv")
df = df.drop_duplicates()

for col in ["region", "payment_mode"]:
                                df[col]       = df[col].astype(str).str.strip().str.title()

for col in ["customer_age", "delivery_days", "rating"]:
                                df[col]       = df[col].fillna(df[col].median())

q1 = df["quantity"].quantile(.25)
q3 = df["quantity"].quantile(.75)
iqr = q3 - q1
upper = q3 + 1.5 * iqr
df["quantity"]                                = df["quantity"].clip(upper=upper)
```

```
df["order_date"]                                =
pd.to_datetime(df["order_date"])
df["gross_amount"]    =    df["quantity"]    *
df["unit_price"]
df["revenue"]  =  df["gross_amount"]  *  (1  -
df["discount"] / 100)
df["month"] = df["order_date"].dt.month
df["weekday"]                                   =
df["order_date"].dt.day_name()
df["is_weekend"]                                =
df["weekday"].isin(["Saturday", "Sunday"])


df.to_csv("retail_sales_2024_cleaned.csv",
index=False)
```

A.2 KPI Calculation

```
metrics = {
    "total_revenue": df["revenue"].sum(),
    "gross_sales": df["gross_amount"].sum(),
    "orders": len(df),
                            "unique_customers":
df["customer_id"].nunique(),
                         "average_order_value":
df["revenue"].mean(),
    "average_rating": df["rating"].mean(),
                        "average_delivery_days":
df["delivery_days"].mean(),
    "average_discount": df["discount"].mean()
```

```
}

print(metrics)
```

A.3 Statistical Checks

```
weekend = df.loc[df["is_weekend"], "revenue"]

weekday = df.loc[~df["is_weekend"], "revenue"]

print(ttest_ind(weekend, weekday, equal_var=False))


print(pearsonr(df["delivery_days"], df["rating"]))
```

APPENDIX B: PROJECT OUTPUT SUMMARY

The following summary can be used as a final verification sheet before submission. It keeps the report and presentation aligned with the same project results.

Item	**Final Project Result**
Project title	Data Analysis and Visualization Using Python
Dataset type	Simulated retail-sales data, 2024
Raw data	5,060 rows and 13 columns
Duplicate rows detected	52
Missing cells	220
Quantity outliers	25
Clean analytical rows	5,008
Total revenue	₹169,384,561.26
Gross sales	₹202,063,565.07
Orders	5,008
Unique customers	1,704
Average order value	₹33,822.80
Average rating	3.97
Average delivery	3.51 days
Average discount	16.33%
Dominant category	Electronics — 76.9% of revenue
Leading payment mode	UPI — 38.5% of orders
Main age segment	26–45 — about two thirds of revenue
Forecast method	Simple linear trend

Forecast period	Next three months

B.1 Final Submission Check

• Insert the original organization completion certificate.

• Replace the registration-number placeholder with the official number.

• Verify the official training dates and mentor name before submission.

• Check that the final PDF remains between 40 and 60 pages.

• Keep the report on A4 paper with the existing LPU-aligned margins and formatting.

• Ensure all figures and tables have correct chapter-based numbering.

• Ensure the presentation and report use the same project title, objectives, results and conclusion.